Type-Dependent Interpolation Instability for Incongruent Text Concepts

Chenhao Tan and NeuriCo

Abstract

Linear concept arithmetic is often evaluated on clean counterfactual pairs, but that setup leaves open a harder question: what happens along the straight-line path between endpoints that are semantically incompatible? We study this question in text embedding space with a two-stage design built on a real embedding model, TEXT-EMBEDDING-3-LARGE. First, we run an LRH-style separable-control experiment on curated word-pair assets and recover a strong linear regime for several transformations, including `Ving-3pSg` and adjective inflection contrasts. Second, we compare matched congruent and incongruent interpolation paths on 225 SUGARCREPE++ triplets spanning object swaps, attribute swaps, and relation replacements. We evaluate each path with off-manifold deviation, dual-space path length, dual detour ratio, switch count, and graph detour ratio.

The aggregate result contradicts a simple "incongruent means more curved" hypothesis: pooled incongruent paths have lower mean off-manifold deviation than congruent controls by 0.0083 and shorter dual paths by 0.0462. The key finding is heterogeneity. Relation replacements behave like genuine interpolation failures, increasing off-manifold deviation by 0.0154 and dual path length by 0.0785 relative to their matched paraphrase controls, while object and attribute swaps often look easier because the negative caption remains lexically close to the source. These results suggest that interpolation instability tracks structural incompatibility, especially disrupted relations, rather than contradiction alone. This narrows when linear traversal is a meaningful tool for representation analysis and latent steering.

1 Introduction

Linear representations are most convincing when they let us move between concepts in a simple and interpretable way. Yet most work on the linear representation hypothesis (LRH) tests whether a concept direction exists, not whether the straight line between two endpoints stays semantically meaningful Park et al. [2024, 2025]. This gap matters in practice. Representation arithmetic, latent traversal, and steering methods all assume that moving in a linear direction will continue to mean something when concepts interact, overlap, or conflict.

Our main question is simple: **what does interpolation look like when the endpoints are incongruent?** Prior LRH work studies curated counterfactual contrasts that are intentionally clean Park et al. [2024, 2025]. Prior interpolation geometry work shows that Euclidean chords can diverge from manifold-respecting paths, but it focuses mostly on generative image latents rather than text representations Shao et al. [2017], Struski et al. [2019], Michelis and Becker [2021]. These two lines of work leave a missing middle ground: pathwise geometry for text endpoints that are meaningful individually but semantically awkward together.

We address this gap with a matched text interpolation study. We first establish a separable-control regime on curated LRH word-pair assets, where the same embedding model recovers strong alignment margins for several concept transformations. We then compare two paths for each SUGAR-

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CREPE++ example: a congruent path from a caption to its paraphrase, and an incongruent path from the same caption to its corrupted alternative. This paired design lets us ask whether incongruence changes path geometry beyond generic manifold sparsity. We measure the answer with off-manifold deviation, graph detour, dual-space path length, and retrieval switching.

The headline result is not a single global penalty for incongruence. At the pooled level, incongruent paths are actually less off-manifold than congruent paths by 0.0083 and shorter in the induced dual space by 0.0462. However, this average hides sharp type dependence. In the `replace_relation` subset, incongruent paths reverse direction and become more unstable than their congruent controls, with a 0.0154 increase in off-manifold deviation and a 0.0785 increase in dual path length. By contrast, object and attribute swaps often create shorter lexical perturbations than the paraphrase baseline, so the incongruent endpoint can sit on an easier local path.

Our contributions. We make four concrete contributions:

- We formulate a matched path-centric evaluation for incongruent text concepts, rather than relying only on pointwise concept directions.
- We conduct a separable-control experiment on LRH assets and show that TEXT-EMBEDDING-3-LARGE still expresses a strong linear regime for several curated transformations.
- We show that interpolation instability is type-dependent: relation replacements behave like true failures, while lexical object and attribute swaps often do not.
- We argue that structural incompatibility is a better operational target than contradiction alone for future interpolation studies and steering analyses.

Paper organization. Section 2 situates the work in concept geometry and interpolation research. Section 3 describes the datasets, metrics, and paired statistical design. Section 4 reports control, pooled, subset, and qualitative results. Section 5 interprets the findings, limitations, and implications.

2 Related Work

Linear representation hypotheses. LRH work asks when model representations admit approximately linear concept structure. Park et al. formalize LRH with counterfactual pairs, concept directions, and a causal inner product that better matches semantic structure than naive Euclidean comparisons Park et al. [2024]. Follow-up work extends this geometry to categorical and hierarchical concepts, showing that richer concept families can still occupy organized linear structure Park et al. [2025]. These papers establish the clean, separable regime that motivates our control experiment. Unlike them, we focus on whether the *path* between endpoints remains meaningful when those endpoints are not cleanly separable.

Ambiguous and non-binary concepts. The assumption of a single clean counterfactual direction becomes harder to maintain for ambiguous or entangled concepts. Nguyen and Leng propose a more flexible maximum-likelihood framework for LRH-style estimation, arguing that realistic concept variation often requires more than a strict paired-difference estimator Nguyen and Leng [2025]. Our findings are compatible with that view: the failure mode we observe is not binary contradiction in the abstract, but changes that disrupt relational structure while keeping surface form nearby.

Interpolation geometry and manifold structure. A separate line of work studies when straight-line interpolation is a poor proxy for geodesic traversal. Shao et al. derive the pull-back geometry of deep generative models and show how to compute curves that respect the learned manifold Shao et al. [2017]. Struski et al. generalize interpolation to feature-aware geodesics under arbitrary latent densities Struski et al. [2019]. Michelis and Becker show that linear interpolation can be arbitrarily suboptimal even when endpoints look reasonable Michelis and Becker [2021]. This literature motivates our path metrics, but it says little about text representations or semantic incongruence. We bring these geometric ideas into a text benchmark with matched congruent and incongruent endpoints.

Positioning. Our work bridges these themes. We inherit the control logic of LRH, but our central object is the interpolation trajectory rather than the direction estimate. We borrow manifold-sensitive evaluation from latent interpolation research, but we apply it to text embeddings and analyze seman-

tic mismatch types directly. The main empirical payoff is a more precise claim: not all incongruence destabilizes interpolation. Relation changes do; short lexical swaps often do not.

3 Methodology

Problem setup. We study straight-line interpolation between text embedding endpoints. Given two texts with embeddings $\boldsymbol{x}_0, \boldsymbol{x}_1 \in \mathbb{R}^d$, we evaluate interior points $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$ on an 11-point grid from $t = 0$ to 1. Our goal is not to build a new interpolation algorithm. Instead, we test whether path behavior changes when one endpoint is a congruent paraphrase and the other is an incongruent alternative.

Model and implementation. All embeddings come from TEXT-EMBEDDING-3-LARGE through the OpenAI embeddings API. We cache responses on disk in `results/cache/` and run the experiment with seed 42. The final run uses Python 3.12.8, 2,000 bootstrap samples, and a k-nearest-neighbor graph with $k = 15$. Although the machine exposes four NVIDIA RTX A6000 GPUs, the final pipeline is API-based and does not use them.

3.1 Datasets and Experimental Design

Separable-control assets. We use 27 curated concept files from `code/linear_rep_geometry/word_pairs/`. Each file contains paired lexical contrasts such as inflectional changes or language mappings. We split pairs into train and test sets, estimate a mean concept vector from training differences, and measure whether held-out differences align more strongly with the correct concept vector than with other concept vectors. This is a sanity check, not a full causal-inner-product replication.

Matched caption triplets. Our main experiment uses 225 examples from SUGARCREPE++, sampled evenly across `swap_object`, `swap_atribute`, and `replace_relation`. Each example supplies a source caption, a congruent paraphrase, and a negative caption. We compare two paths for the same source:

$$\text{congruent path}: \quad \texttt{caption} \rightarrow \texttt{caption2}, \tag{1}$$

$$\text{incongruent path}: \quad \texttt{caption} \rightarrow \texttt{negative_caption}. \tag{2}$$

This matched design controls for image identity and much of the local lexical context.

3.2 Metrics

Control metric. For the LRH assets, we report same-concept alignment, off-target alignment, and their difference as an alignment margin. Larger margins indicate stronger separability with less cross-concept interference.

Path geometry metrics. For SUGARCREPE++ paths we compute five metrics.

- **off-manifold deviation:** average nearest-neighbor distance from interior interpolation points to the sampled caption set.
- **dual path length:** cumulative Jensen-Shannon length after projecting each interpolation point into a local similarity-softmax simplex.
- **dual detour ratio:** dual path length divided by the endpoint distance in the same dual space.
- **switch count:** number of changes in source/target/alternate anchor assignment along the path.
- **graph detour ratio:** shortest-path distance on the k-NN graph divided by direct endpoint distance.

Baselines. We compare three baselines. First, other concept vectors act as off-target baselines in the separable-control experiment. Second, congruent paraphrase interpolation is the primary baseline for incongruent interpolation. Third, direct endpoint distance is the denominator baseline for graph and dual detour ratios.

3.3 Statistical Analysis

Paired inference. We evaluate congruent versus incongruent differences with a paired permutation sign test. We report 95% bootstrap confidence intervals for mean differences, paired standardized

Table 1: Top separable concepts recovered in the LRH-style control experiment. Higher same-concept alignment and larger margins are better; lower off-target alignment indicates less interference.

Concept	Same-concept	Off-target	Margin
`Ving - 3pSg`	0.629	-0.018	**0.647**
`adj - superlative`	**0.668**	0.052	0.616
`adj - comparative`	0.620	0.033	0.587
`3pSg - Ved`	0.568	-0.000	0.568
`verb - Ving`	0.587	0.061	0.525

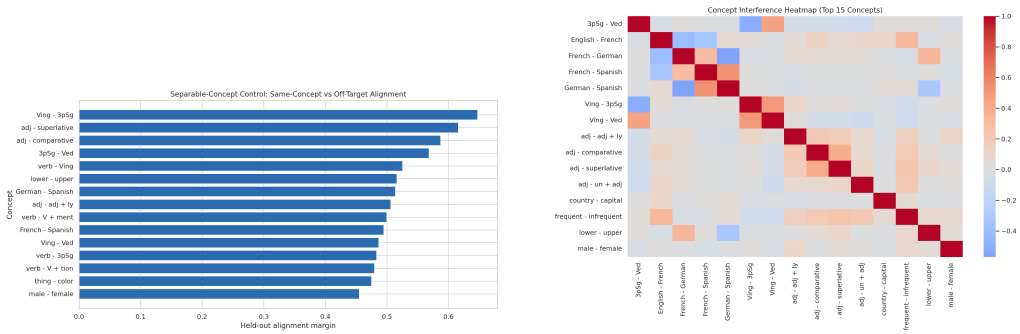

Figure 1: Control diagnostics on curated LRH word-pair assets. Left: top alignment margins for held-out concept pairs. Right: cross-concept interference matrix. Strong margins and sparse interference show that the embedding model still exhibits a clean linear regime on curated contrasts.

effect sizes (d_z), and Benjamini-Hochberg corrected q-values across the main metric family. Because the design is paired within each example, the key comparisons are within-example differences rather than cross-example averages alone.

4 Results

Separable controls recover a linear regime. The control experiment confirms that TEXT-EMBEDDING-3-LARGE is not uniformly noisy. The best concepts show large margins with near-zero or negative off-target alignment. For example, `Ving-3pSg` reaches a margin of 0.647, and adjective superlatives and comparatives exceed 0.58. Figure 1 visualizes the strongest recovered margins, while figure 1 shows that interference is limited and concentrated in morphologically related concepts such as verb tense variants.

Pooled incongruent paths do not look globally worse. At the pooled level, incongruent paths are *less* off-manifold and shorter in the dual space than congruent controls. The mean differences are small but statistically reliable for off-manifold deviation (-0.0083, $q = 0.0262$) and dual path length (-0.0462, $q = 0.0262$). By contrast, dual detour ratio and switch count do not survive correction, and graph detour ratio collapses to 1.0 for both conditions. This pooled result directly contradicts the original simple hypothesis that incongruence should always produce a more curved or unstable path.

Subset heterogeneity drives the pooled reversal. The reversal is not noise; it comes from clear heterogeneity across corruption types. The `replace_relation` subset behaves like genuine incongruence: incongruent endpoints are farther from the source than paraphrases, and the interpolation path becomes both more off-manifold and longer in the dual space. By contrast, in `swap_atribute` and `swap_object`, the negative endpoint is often *closer* to the source than the paraphrase. This makes the incongruent path locally easier even when the sentence is semantically corrupted.

Qualitative trajectories isolate the failure mode. The retrieval trajectories show what "instability" means in practice. In `replace_relation`, the midpoint often flips abruptly between incompatible event descriptions rather than moving through a smooth semantic neighborhood. One representative

Metric	Congruent mean	Incongruent mean	Δ (inc $-$ cong)	95% CI	p	q
Off-manifold deviation	0.1325	**0.1242**	**-0.0083**	[-0.0143, -0.0020]	0.0105	0.0262
Dual path length	0.5108	**0.4645**	**-0.0462**	[-0.0777, -0.0129]	0.0075	0.0262
Dual detour ratio	1.0612	**1.0567**	-0.0045	[-0.0133, 0.0047]	0.3193	0.3992
Switch count	1.0133	**1.0000**	-0.0133	[-0.0311, 0.0000]	0.2654	0.3992
Graph detour ratio	1.0000	1.0000	0.0000	[0.0000, 0.0000]	1.0000	1.0000

Table 2: Paired comparison on 225 SUGARCREPE++ examples. Best values in each row are shown in **bold**, taking lower values as better because all metrics measure deviation or path length.

Table 3: Subset breakdown of the main effect. Positive differences mean incongruent paths are more unstable than congruent controls. Direct distances explain why lexical swaps often reverse the pooled hypothesis.

Subset	Δ off-manifold	Δ dual path	Δ dual detour	Congruent dist.	Incongruent dist.
`replace_relation`	**+0.0154**	**+0.0785**	**+0.0280**	0.4876	**0.5429**
`swap_atribute`	-0.0235	-0.1316	-0.0199	**0.4672**	0.3826
`swap_object`	-0.0167	-0.0857	-0.0216	**0.4762**	0.4161

source, "Two people are holding phones together to create a linking," shifts toward the contradictory target "Two men are dropping phones, causing them to unlink," with off-manifold deviation of 0.2706 and dual detour ratio of 1.3243. Another source, "The back view of an airplane on a runway," transitions toward "An airplane is flying over a runway," with off-manifold deviation 0.2672 and dual detour ratio 1.3323. These are the clearest examples of a straight chord leaving the empirical semantic manifold.

By contrast, some `swap_object` negatives remain locally close to the source because they preserve most of the wording. For example, the source "A city scape scene" paired with the negative "A train yard scene with a city scape in the foreground" yields a lower off-manifold deviation score (0.2464) than the paraphrase control (0.2660), even though the negative is semantically wrong. This explains why contradiction alone is too coarse a predictor of path instability.

5 Discussion

What do the results mean? The main empirical result is a qualified answer to the motivating question. Interpolation between incongruent concepts does not have one generic signature. Instead, instability depends on the kind of mismatch that defines the endpoint pair. When the mismatch changes relational structure, the path behaves more like the geodesic failures described in latent-space geometry work Shao et al. [2017], Michelis and Becker [2021]. When the mismatch is a short lexical substitution, the negative endpoint can remain close enough to the source that the linear path still looks locally regular.

Why does pooled incongruence look easier? The pooled reversal comes from benchmark construction, not from evidence against path sensitivity. In two of the three subsets, the negative caption is often lexically closer to the source than the paraphrase. That makes the incongruent endpoint a shorter chord in embedding space. A shorter chord naturally reduces nearest-neighbor deviation and dual-space travel, even if the endpoint is less truthful. The result is a reminder that truth-conditional corruption and geometric difficulty are not the same variable.

Broader implications. These findings narrow when linear traversal is a useful analysis tool. If a representation change preserves local lexical scaffolding, a straight path may remain interpretable even when the endpoint is wrong. But if the change disrupts who did what to whom, path-based methods become less reliable. This matters for activation steering, latent editing, and causal analysis, where a direction can look clean at the endpoints while the intermediate states are semantically unstable.

Limitations. Our study uses embedding-space geometry rather than hidden-state geometry with the causal inner product of Park et al. Park et al. [2024]. The results therefore test a broader LRH-inspired hypothesis rather than the full formalism. The graph detour metric is also weak in this setting because the sampled k-NN graph produces a detour ratio of 1.0 for every example, leaving off-

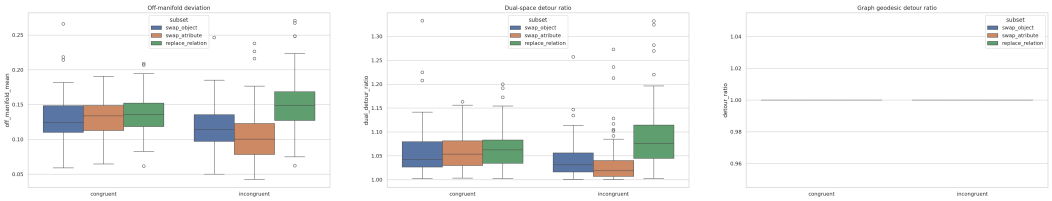

Figure 2: Main path metrics by subset and pair type. Left: off-manifold deviation shows the strongest reversal between relation replacements and lexical swaps. Center: dual detour is modestly higher for relation replacements but close elsewhere. Right: graph detour ratio is uninformative in this sampled graph because both conditions remain on shortest paths of length equal to direct distance.

manifold and dual-space measures as the informative diagnostics. Finally, SUGARCREPE++ is only a proxy for incongruence. Its subsets mix distinct corruption types, and the original `swap_atribute` split is orthographically inconsistent and semantically closer to lexical editing than to structural contradiction.

6 Conclusion

We studied interpolation for incongruent text concepts with a paired, path-centric evaluation. A control experiment on curated LRH assets showed that TEXT-EMBEDDING-3-LARGE still supports strong linear contrasts in separable settings. The main experiment then showed that incongruence does not produce a universal interpolation penalty. At the pooled level, incongruent paths were slightly shorter and less off-manifold than congruent paraphrase controls. The decisive result is subset-specific: relation replacements increase off-manifold deviation and dual-space path length, while object and attribute swaps often do not because they remain lexically close to the source.

The key takeaway is that interpolation failure tracks structural incompatibility more than contradiction alone. Future work should test the same question in hidden-state spaces with explicit LRH estimators, use benchmarks whose negative endpoints are not systematically closer than paraphrases, and adopt stronger manifold estimators so geodesic detour becomes informative instead of degenerate.

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